

Sedative and Hypnotic Drugs

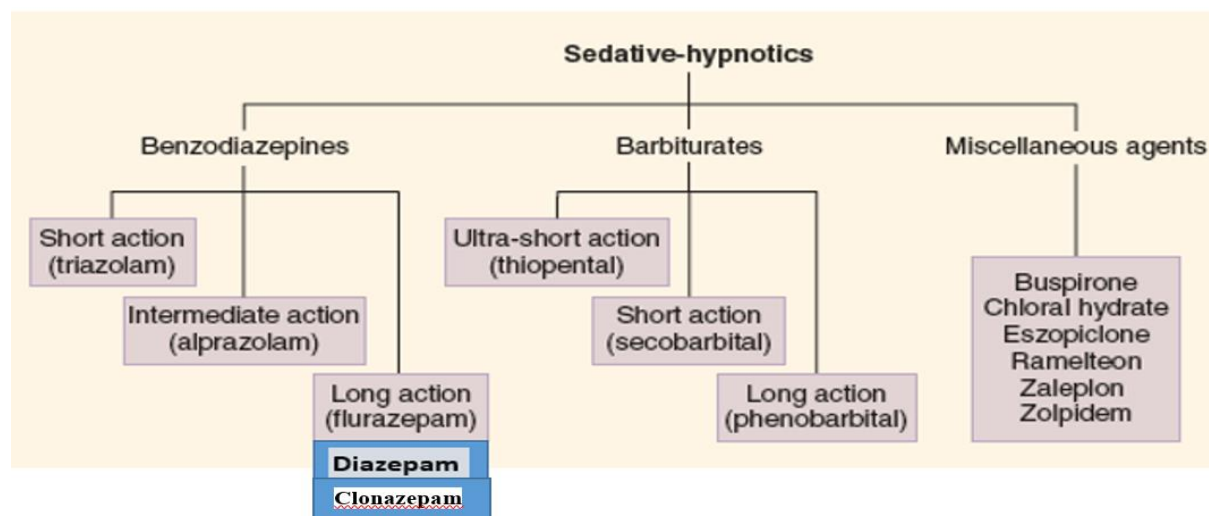
An effective sedative (anxiolytic) drug: agent should reduce anxiety and exert a calming effect.

An effective hypnotic drug: agent should produce drowsiness and encourage the onset and maintenance of a state of sleep.

Hypnotic effects involve more pronounced depression of the CNS than sedation.

Graded dose-dependent depression of CNS function is a characteristic of most sedative-hypnotics.

Classification of sedative and hypnotic drugs:

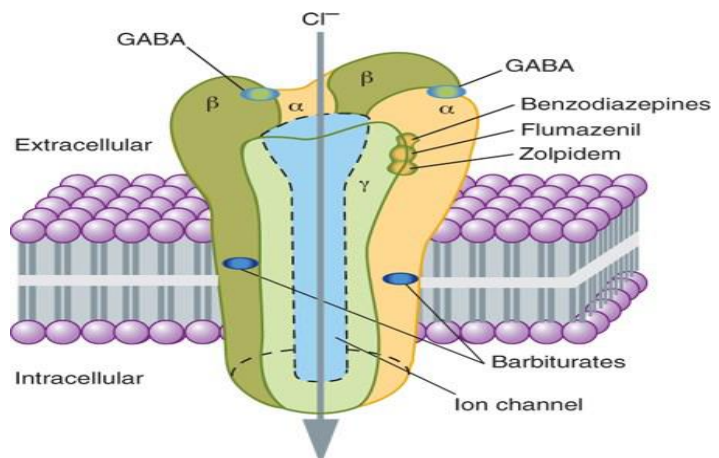
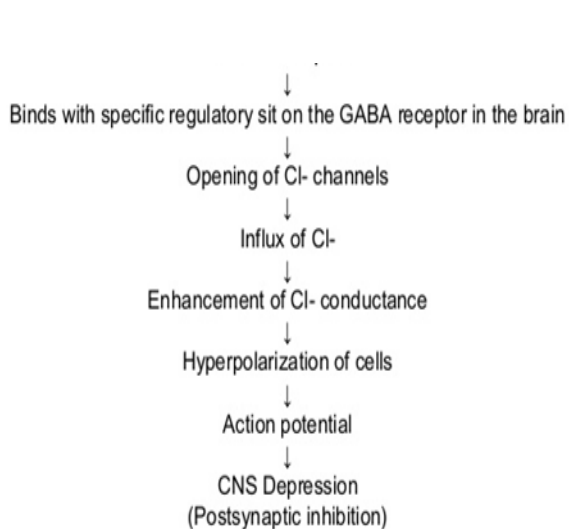


Mechanism of Action:

- The benzodiazepines, the barbiturates, zolpidem, zaleplon, eszopiclone, and many other drugs bind to molecular components of the GABA_A receptor in neuronal membranes in the CNS.
- This receptor, which functions as a chloride ion channel, is activated by the inhibitory neurotransmitter GABA (Figure 1). The GABA_A receptor has a pentameric structure assembled from five subunits.
- Multiple forms of α , β , and γ subunits are arranged in different pentameric combinations so that GABA_A receptors exhibit **molecular heterogeneity**.

GABA appears to interact at two sites between α and β subunits triggering chloride channel opening with resulting membrane hyperpolarization.

- Binding of benzodiazepines and the newer hypnotic drugs such as zolpidem occurs at a single site between α and γ subunits, facilitating the process of chloride ion channel opening.
- Note that these binding sites are distinct from those of the barbiturates. A major isoform of the GABA_A receptor that is found in many regions of the brain consists of two α 1 subunits, two β 2 subunits, and one γ 2 subunit.
- In contrast to benzodiazepines; zolpidem, zaleplon, and eszopiclone bind more selectively because these drugs interact only with GABA_A-receptor isoforms that contain α 1 subunits so they are only *hypnotic drugs*.
- The heterogeneity of GABA_A receptors may constitute the molecular basis for the varied pharmacologic actions of benzodiazepines and related drugs.
- In contrast to GABA itself, benzodiazepines and other sedative-hypnotics have a low affinity for GABA_B receptors, which are activated by the spasmolytic drug baclofen.



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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Figure (1): Mechanism of sedative hypnotic action & GABA_A receptor, showing the chloride ionic channel formed of α , β , and γ subunits, GABA neurotransmitter binding sites, and the sedative hypnotic binding sites.

- The benzodiazepines do not substitute for GABA but appear to enhance GABA's effects allosterically without directly activating GABA_A receptors or opening the associated chloride channels. This enhancement takes the form of an **increase in the frequency of channel-opening**.
- Barbiturates also facilitate the actions of GABA at multiple sites in the CNS, but—in contrast to benzodiazepines—they appear to **increase the duration of the GABA-gated chloride channel openings**. Barbiturates are less selective in their actions than benzodiazepines, because they also depress the actions of the excitatory neurotransmitter glutamic acid via binding to the AMPA receptor so it has low safety margin.
- Anesthetic concentrations of pentobarbital also block high frequency sodium channels. This multiplicity of sites of action of barbiturates may be the basis for their ability to induce full surgical anesthesia and for their more pronounced central depressant effects compared with benzodiazepines and the newer hypnotics.
- Barbiturates use is associated with rapid development of tolerance and physical dependence with subsequent withdrawal symptoms. It also lack specific antidote.
- Thiopental is used as a general anesthetic and phenobarbital is used as anticonvulsant.

Pharmacokinetics

A . Absorption and Distribution

- **Lipid solubility** plays a major role in determining the rate at which a particular sedative-hypnotic enters the CNS. This property is responsible for the **rapid onset** of the effects of triazolam, thiopental, and the newer hypnotics.
- All sedative-hypnotics cross the placental barrier during pregnancy. If sedative-hypnotics are given during the predelivery period, they may contribute to the depression of neonatal vital functions. Sedative-hypnotics are also detectable in breast milk and may exert depressant effects in the nursing infant.

B . Biotransformation

- Metabolic transformation to more water-soluble metabolites is necessary for clearance of sedative-hypnotics from the body. The microsomal drug metabolizing enzyme systems of the liver are most important in this regard, so **elimination half-life of these drugs depends mainly on the rate of their metabolic transformation.**
- Hepatic metabolism accounts for the clearance of all benzodiazepines. The patterns and rates of metabolism depend on the individual drugs.
- Most benzodiazepines undergo microsomal oxidation (phase I reactions) catalyzed by cytochrome P450 isozymes. The metabolites are subsequently conjugated (phase II reactions) to form glucuronides that are excreted in the urine.
- However, many phase I metabolites of benzodiazepines are *pharmacologically active*, some with long half-lives. For example, **desmethyldiazepam**, which has an elimination half-life of more than 40 hours, is an active metabolite of, diazepam.

- Alprazolam and triazolam undergo α -hydroxylation, and the resulting metabolites appear to exert short-lived pharmacologic effects because they are rapidly conjugated to form inactive glucuronides.
- **The short elimination half-life of triazolam (2–3 hours) favors its use as a hypnotic rather than as a sedative drug.**
- With exception of **phenobarbital** (not metabolized) the liver metabolizes all barbiturates by oxidation followed by glucourinidation. Most of barbiturates have long elimination half-lives so repeated doses may lead to cumulative effect.
- Most of the newer agents as zolpidem, Zaleplon, eszopiclone, suvorexant are also metabolized by CYP450 so also affected by inducers or inhibitors.

C. Excretion

The water-soluble metabolites of sedative-hypnotics, formed mostly via the phase II conjugation, are excreted mainly via the kidney. In most cases, changes in renal function do not have a marked effect on the elimination of parent drugs.

Phenobarbital is excreted unchanged

Effects:

1. Sedation

- Benzodiazepines, barbiturates, and most older sedative-hypnotic drugs exert calming effects with concomitant reduction of anxiety at relatively low doses. In most cases, however, the anxiolytic actions of sedative-hypnotics are accompanied by some depressant effects on psychomotor and cognitive functions.
- The benzodiazepines also exert dose-dependent **anterograde amnesic effects** (inability to remember events occurring during the drug's duration of action).

2. Hypnosis

By definition, all of the sedative-hypnotics induce sleep if high enough doses are given. The latency of sleep onset is decreased (time to fall asleep) and the duration of sleep is increased. The use of sedative-hypnotics for more than 1–2 weeks leads to some tolerance to their effects on sleep patterns.

3. Anesthesia

- High doses of certain sedative-hypnotics depress the CNS to the point known as stage III of general anesthesia. Among the barbiturates, **thiopental** is very lipid-soluble, penetrating brain tissue rapidly following intravenous administration, a characteristic favoring their use for the induction of anesthesia.
- Rapid tissue **redistribution** (not rapid elimination) accounts for the short duration of action of these drugs, a feature useful in recovery from anesthesia.
- Benzodiazepines—including diazepam, and midazolam—are used intravenously in anesthesia often in combination with other agents.

4. Anti-convulsant effects

- Most sedative-hypnotics are capable of inhibiting the development and spread of epileptiform electrical activity in the CNS
- Several benzodiazepines—including clonazepam and diazepam are sufficiently selective to be clinically useful in the management of seizures.
- Of the barbiturates, phenobarbital is effective in the treatment of generalized tonic-clonic seizures, though not the drugs of first choice. However, zolpidem, zaleplon, and eszopiclone lack anticonvulsant activity.

5- Anti-spasticity:

At high doses, the benzodiazepines relax the spasticity of skeletal muscle, probably by increasing presynaptic inhibition in the spinal cord.

6- Effects on respiration and cardiovascular function

- At hypnotic doses in *healthy patients*, the effects of sedative-hypnotics on respiration are comparable to changes during natural sleep.
- However, even at therapeutic doses, sedative-hypnotics can produce significant ***respiratory depression*** in patients with ***pulmonary disease***.
- Effects on respiration are dose-related, and depression of the medullary respiratory center is the usual cause of death due to overdose of sedative-hypnotics.
- At doses up to those causing hypnosis, ***no significant effects on the cardiovascular system*** are observed in *healthy patients*. However, in ***hypovolemic states, heart failure***, the normal doses of sedative-hypnotics may cause ***cardiovascular depression***,
- Respiratory and cardiovascular effects are more marked when sedative-hypnotics are given intravenously.

Tolerance—decreased responsiveness to a drug following repeated exposure—is a common feature of sedative-hypnotic use. It is important to recognize that partial cross tolerance

occurs between the sedative-hypnotics described here and also with ethanol

Physiologic dependence and abstinence or withdrawal syndrome

- Can occur with prolonged use of sedative hypnotics and sudden cessation. This syndrome is characterized by states of increased anxiety, insomnia, and CNS excitability that may progress to convulsions.
- Differences in the severity of withdrawal symptoms resulting from individual sedative-hypnotics relate in part to half-life, since drugs with long half-lives are eliminated slowly enough to accomplish gradual withdrawal with few physical symptoms.
- The abrupt cessation of zolpidem, zaleplon, or eszopiclone also may result in withdrawal symptoms, though usually of less intensity than those seen with benzodiazepines.

BENZODIAZEPINE ANTAGONISTS: FLUMAZENIL

- Flumazenil binds with high affinity for the benzodiazepine binding site on the GABA_A receptor that act as competitive antagonists. It blocks many of the actions of benzodiazepines, zolpidem, zaleplon, and eszopiclone.
- Flumazenil is approved for use in reversing the CNS depressant effects of benzodiazepine when given **intravenously**.
- Flumazenil acts rapidly but has a short half-life (0.7–1.3 hours) due to rapid hepatic clearance. Because all benzodiazepines have a longer duration of action than flumazenil, sedation commonly recurs, requiring repeated administration of the antagonist.

Adverse effects of flumazenil include dizziness, nausea, vomiting, and agitation,.....

Clinical uses of sedative-hypnotics.

- 1- For relief of anxiety
- 2- For insomnia
- 3- For sedation and amnesia before and during medical and surgical procedures
- 4- For treatment of epilepsy and seizure states
- 5- As a component of balanced anesthesia (intravenous administration)
- 6- For control of ethanol or other sedative-hypnotic withdrawal states
- 7- For muscle relaxation in specific neuromuscular disorders

Side effects

A. Psychomotor Dysfunction

- This includes cognitive impairment, decreased psychomotor skills, and unwanted daytime sedation. These effects are particularly dangerous when driving or operating other machinery. These adverse effects are more common with

benzodiazepines that have active metabolites with long half-lives (eg, diazepam, flurazepam), but can also occur after a single dose of a short-acting benzodiazepine such as triazolam.

- The dosage of a sedative-hypnotic should be reduced in *elderly* patients, who are more susceptible to drugs that cause psychomotor dysfunction. In such patients excessive daytime sedation has been shown to increase the risk of falls and fractures.

B. Anterograde amnesia

This occurs with benzodiazepines, especially when used at high dosage. .”

Zolpidem and the newer hypnotics cause modest day-after psychomotor depression with few amnestic effects.

B. Additive CNS Depression

Additive depression occurs when sedative-hypnotics are used with other drugs in the class as well as with alcoholic beverages, antihistamines, antipsychotic drugs, opioid analgesics, and tricyclic antidepressants. This is the most common type of drug interaction involving sedative-hypnotics.

C. Overdosage

- Overdosage of sedative-hypnotic drugs causes severe respiratory and cardiovascular depression; these potentially lethal effects are more likely to occur with alcohols and barbiturates, than with benzodiazepines or the newer hypnotics such as zolpidem.

- Management of intoxication requires maintenance of a patent airway and ventilatory support. Flumazenil may reverse CNS depressant effects of benzodiazepines, eszopiclone, zolpidem, and zaleplon but has no beneficial actions in overdosage with other sedative-hypnotics.

D. Other Adverse Effects

Barbiturates (but not benzodiazepines, eszopiclone, zolpidem, or zaleplon) induce the formation of the liver microsomal enzymes that metabolize drugs. This enzyme induction may lead to multiple drug interactions.

ATYPICAL SEDATIVE-HYPNOTICS

A. Buspirone

- Buspirone is a selective anxiolytic, with minimal CNS depressant effects (it does not affect driving skills) and has no anticonvulsant or muscle relaxant properties. The drug interacts with the 5-HT_{1A} subclass of brain serotonin receptors as a partial agonist, but the precise mechanism of its anxiolytic effect is unknown. Buspirone has a slow onset of action (>1 week) and is used in generalized anxiety disorders, but is less effective in panic disorders. Tolerance development is minimal with chronic use, and there is little rebound anxiety or withdrawal symptoms on discontinuance.

B. Melatonin Receptor Agonists

Ramelteon

- A novel hypnotic drug prescribed specifically for patients who have difficulty in falling asleep, is an agonist at MT₁ and MT₂ melatonin receptors in the brain.
- Melatonin receptors are thought to be involved in maintaining circadian rhythms underlying the sleep-wake cycle.

C. Orexin Antagonists (Suvorexant)

Orexin is a peptide found in the hypothalamus and is involved in wakefulness. Suvorexant, an antagonist at OX1R and OX2R orexin receptors, has hypnotic properties and is approved for the treatment of insomnia.